



Defining Colors in App Design

Assoc. Prof. Dr. Kanda Runapongsa Saikaew
Computer Engineering Department
Khon Kaen University



Agenda

- Color usage and palettes
- Color theme creation
- Applying color to UI
- Defining colors in your design system



Color usage and palettes

- The Material Design color system helps you apply color to your UI in a meaningful way
- In the color system, you select a primary and a secondary color to represent your brand
- Dark and light variants of each color can then be applied to your UI in different ways

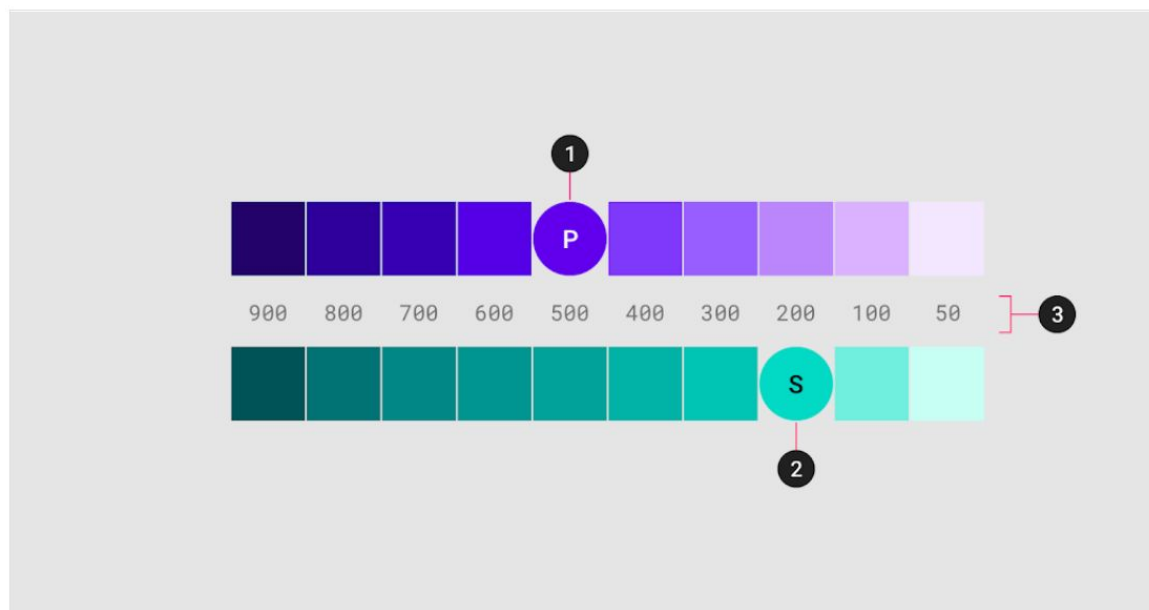


Colors and theming

- Color themes are designed to be harmonious, ensure accessible text, and distinguish UI elements and surfaces from one another
- [The Material Design palette tool](#) or 2014 Material Design palettes are available to help you select colors



The primary color and the secondary color



A sample primary and secondary palette

1. Primary color
2. Secondary color
3. Light and dark variants

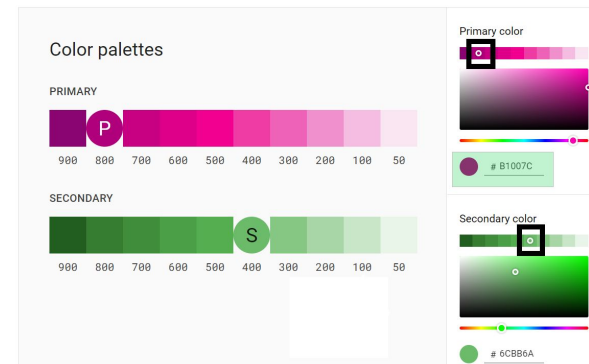
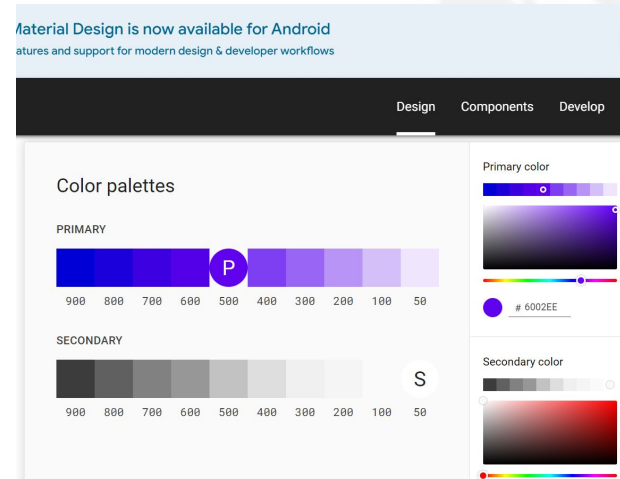
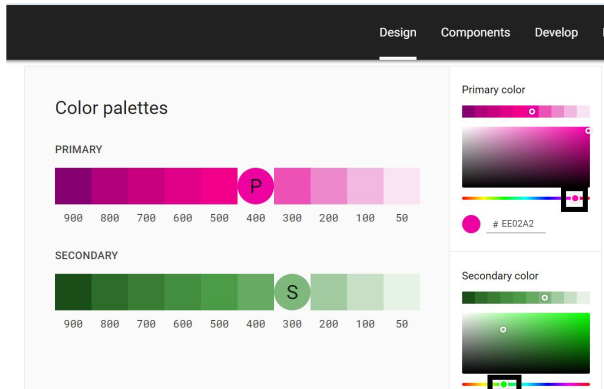
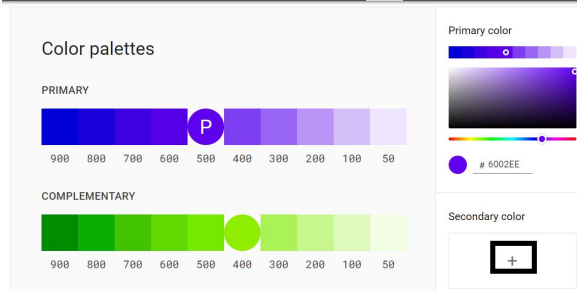
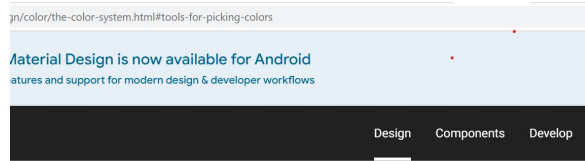


Material design palette

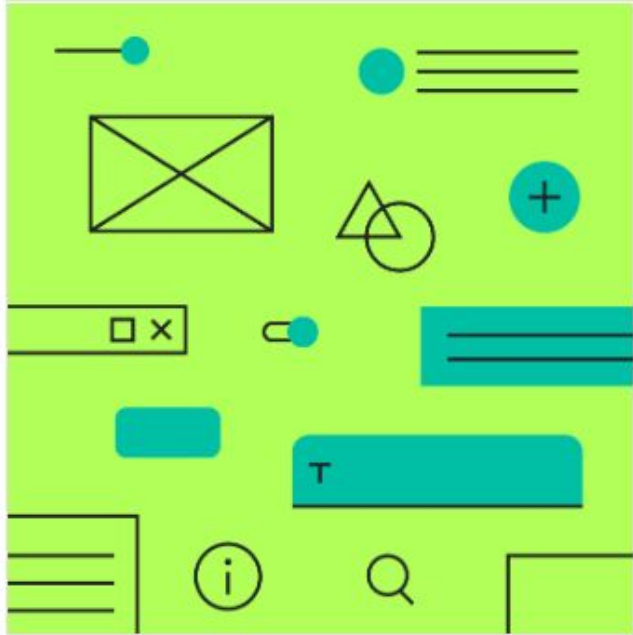
- The Material palette generator can be used to generate a palette for any color you input
- Hue, chroma, and lightness are adjusted by an algorithm that creates palettes that are usable and aesthetically pleasing
- Color palettes can be generated based on the primary input color, and whether the desired palette should be analogous, complementary, or triadic in relation to the primary color.
- Alternatively, the tool can generate expanded palettes, based on any primary and secondary color.



Using the material design palette

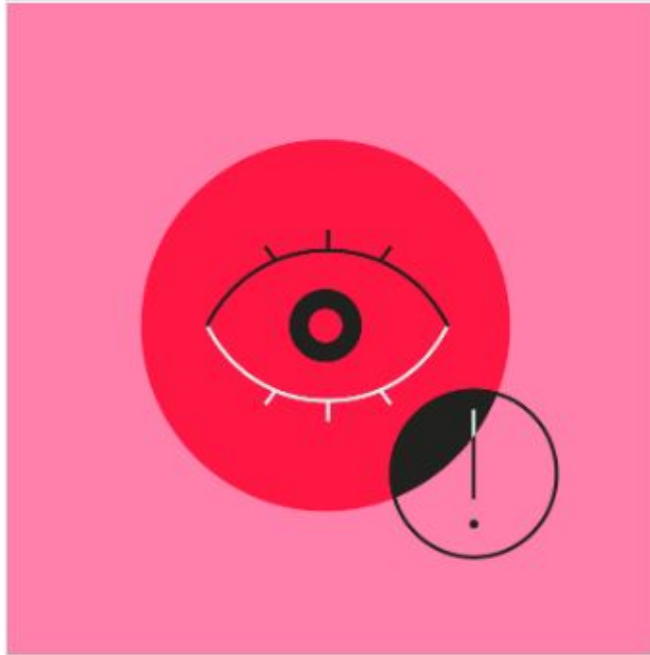


Principles: Hierarchical



- Color indicates
 - Which elements are interactive
 - How they relate to other elements
 - Their level of prominence
- Important elements should stand out the most

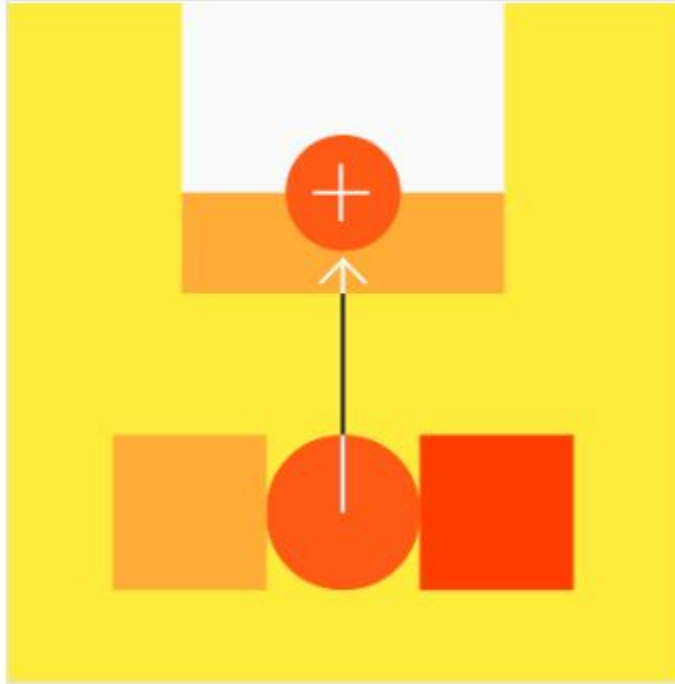
Principles: Legible



- Text and important elements, like icons, should meet legibility standards when appearing on colored backgrounds



Principles: Expressive

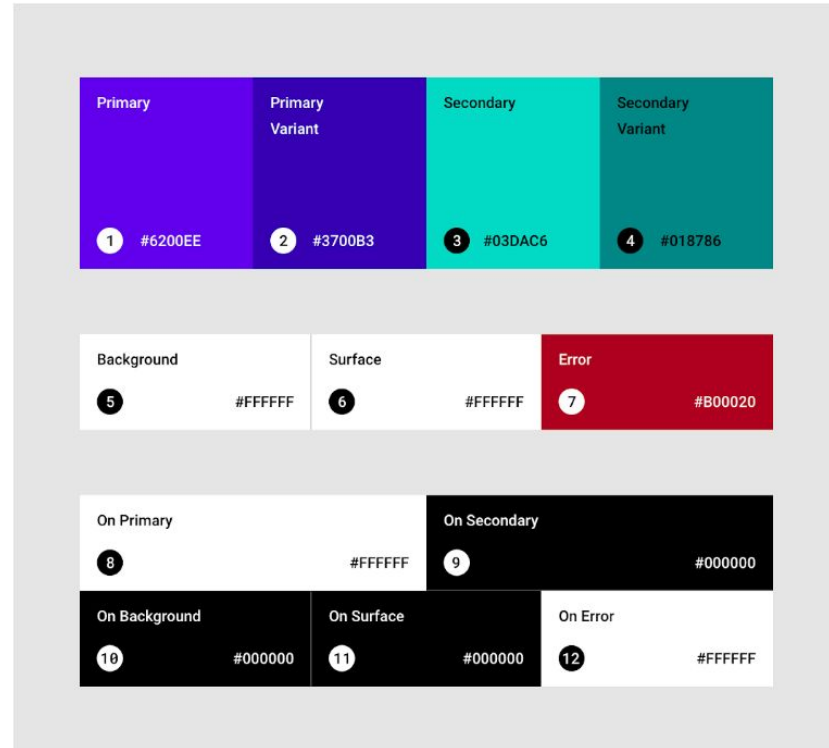


- Show brand colors at memorable moments that reinforce your brand's style



The baseline Material color theme

- This includes default colors for:
 - Primary and secondary colors
 - Variants of primary and secondary colors
 - Additional UI colors, such as colors for backgrounds, surfaces, errors, typography, and iconography



The baseline Material color theme



Primary color (1/2)

- A primary color is the color displayed most frequently across your app's screens and components
- Your primary color can be used to make a color theme for your app, including dark and light primary color variants

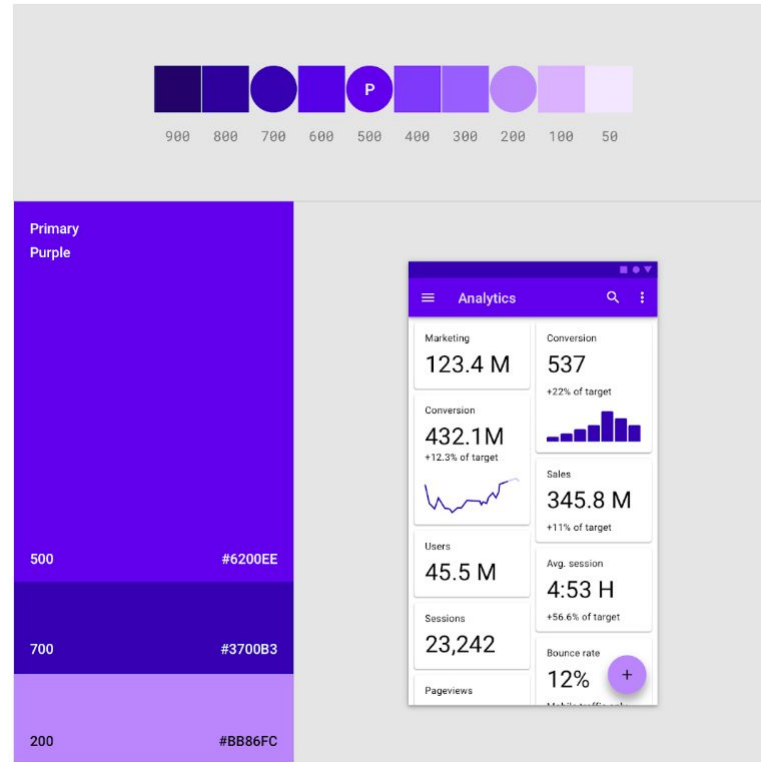


Primary color (2/2)

- Distinguish UI elements
 - To create contrast between UI elements, such as a top app bar from a system bar, you can use light or dark variants of your primary colors
 - You can also use these to distinguish elements within a component, such as the icon of a floating action button from its circular container



A primary color and two primary variants



This UI uses a primary color and two primary variants.

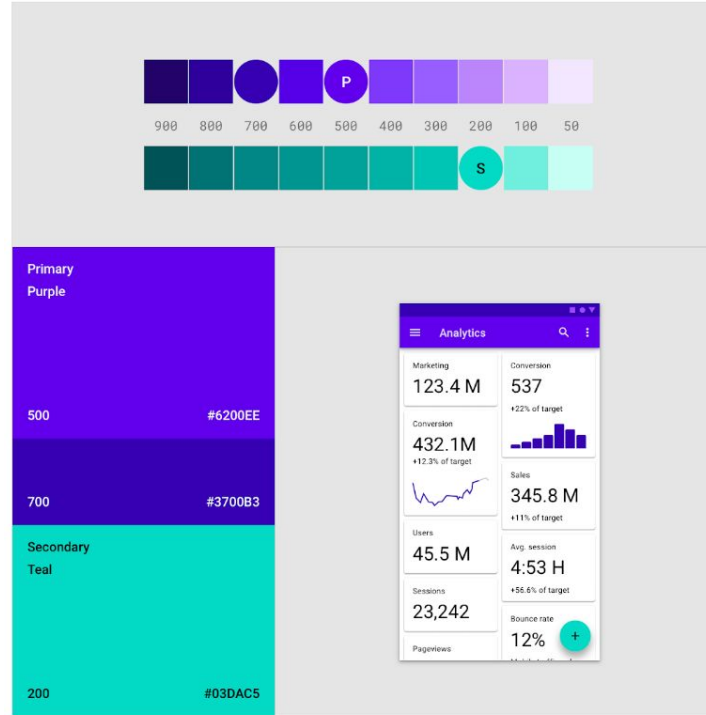


Secondary color

- A secondary color provides more ways to accent and distinguish your product
- If you don't have a secondary color, your primary color can also be used to accent elements
- Secondary colors are best for
 - Floating action buttons
 - Selection controls, like sliders and switches
 - Highlighting selected text
 - Progress bars
 - Links and headline



A primary color, a primary variant, and a secondary color

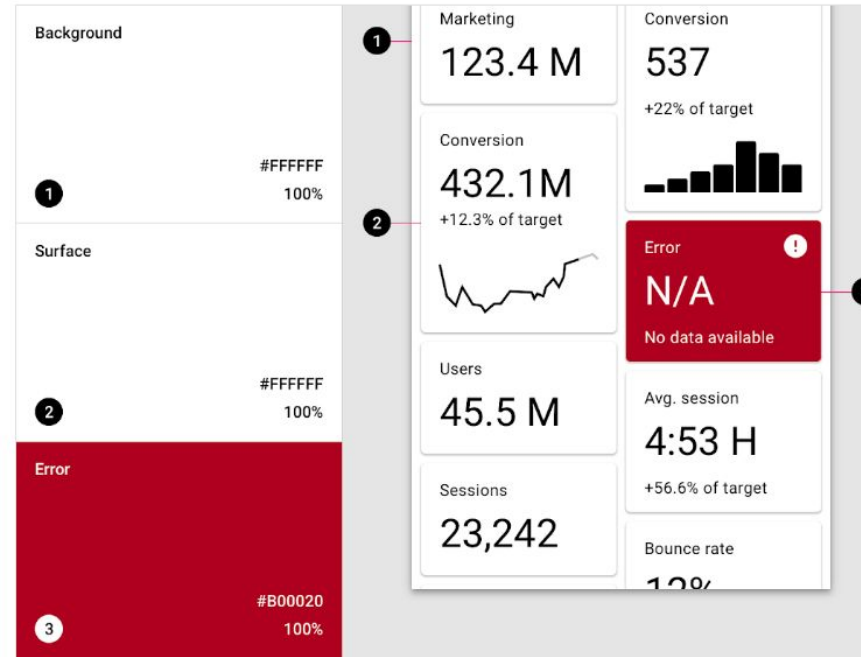


This UI uses a color theme with a primary color, a primary variant, and a secondary color.



Surface, background, and error colors

- Surface, background, and error colors typical don't represent brand



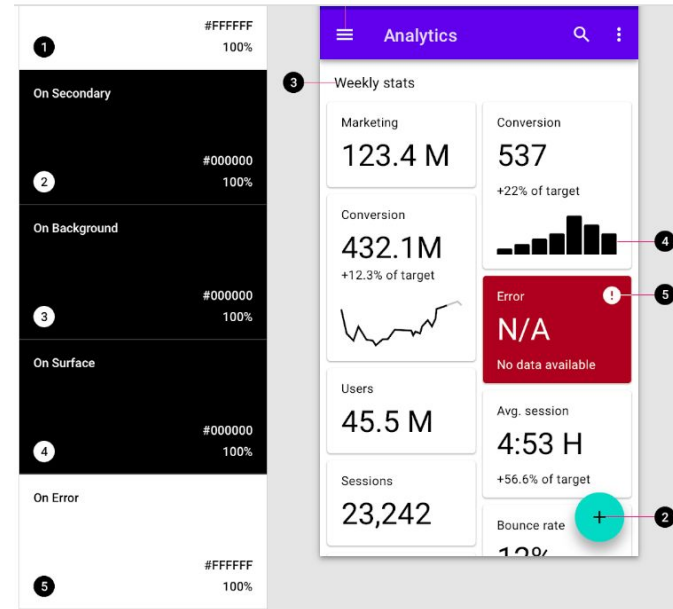
A UI showcasing the baseline colors for background, surface, and error color:

1. Baseline background color: #FFFFFF
2. Baseline surface color: #FFFFFF
3. Baseline error color: #B00020



Typography and iconography colors

- “On” colors
 - They color elements that appear “on” top of surfaces that use the following colors: a primary, secondary color, surface color, background color, or error color



A UI showcases the baseline colors for text and iconography:

1. Baseline on primary color #FFFFFF
2. Baseline on secondary color #FFFFFF
3. Baseline on background color #B00020
4. Baseline on surface color #B00020
5. Baseline on error color #B00020



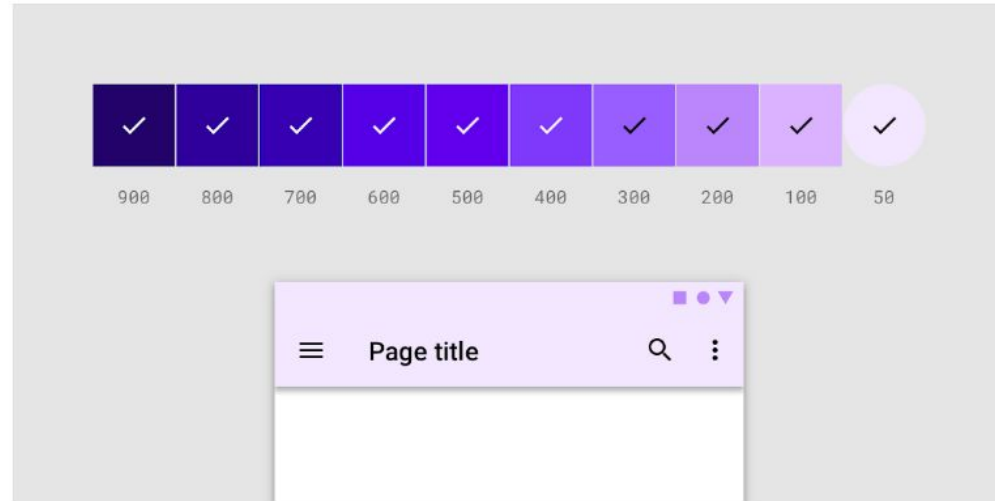
Color swatches

- A swatch is a sample of a color chosen from a range of similar colors



Check marks

- A white (black) check mark indicates when white (black) text is legible on a background color



The 60-30-10 Rule

- A proportion heuristic borrowed from interior design
- ~60% dominant color — usually a neutral (background, surfaces)
- ~30% secondary color — major components, structure
- ~10% accent color — draws the eye (CTA, links, highlights)
- Purpose: prevent a UI that feels monotonous or chaotic
- A *guideline*, not a specification — treat percentages as approximate



Material Design Color System vs 60-30-10 Rule

- Two different layers, not two choices — use both together
- Material system = *which colors exist and what each means*
- 60-30-10 = *how much screen area each color gets*
- Material covers: naming, tonal shades, semantic roles, accessibility
- 60-30-10 covers: visual balance (one dominates, one supports, one accents)



How the Two Map Onto Each Other

60-30-10

Typical

- 60% dominant
 - 30% secondary
 - 10% accent
 - Neutrals / background / surface
 - Primary color
 - Secondary / accent color
-
- Material's "primary" $\neq$ the 60-30-10 dominant — it sits in the 30% band
 - Do not memorize the rows; understand *why* the names diverge



Where They Contrast

- 60-30-10 is silent on accessibility, contrast ratios, and error/disabled states — Material covers these
- Material is silent on proportion — a valid palette can still look unbalanced
- Material 3 dynamic color (from user wallpaper) undercuts fixed ratios → design *tolerances*, not exact percentages
- Neither replaces the other; each fills the other's gap



Light and dark themes

- Some apps have both light and dark themes
- To maintain visibility of elements and legibility of text, you can adapt the different color schemes for dark and light themes



A news app in a light theme uses a primary and secondary scheme.



The same news app in dark theme uses a different color scheme to maintain legibility.



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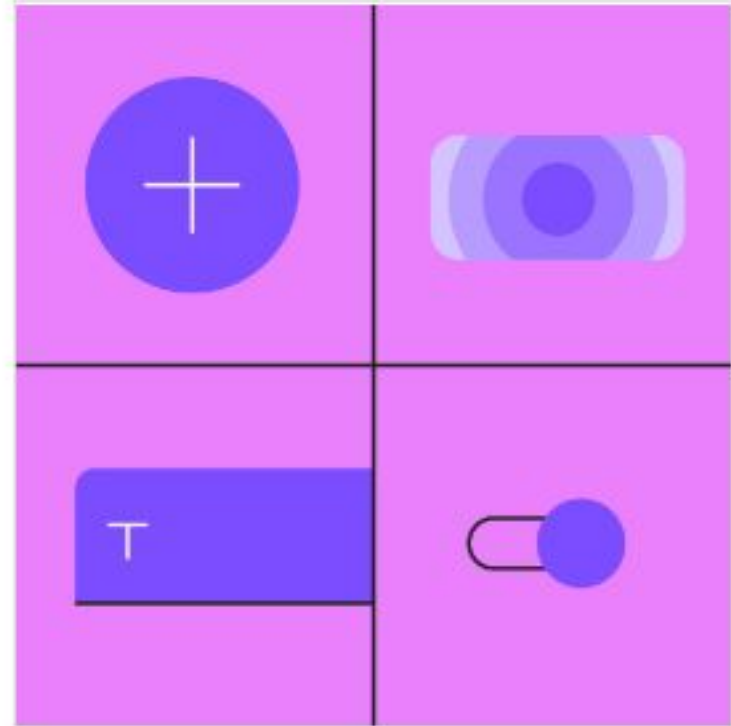
Principles: Consistent

- Color should be applied throughout a UI consistently and be compatible with the brand it presents



Principles: Distinct

- Color should create distinction between elements, with sufficient contrast between them



Principles: Intentional

- Color should be applied purposefully as it can convey meaning in multiple ways, such as relationship between elements and degrees of hierarchy



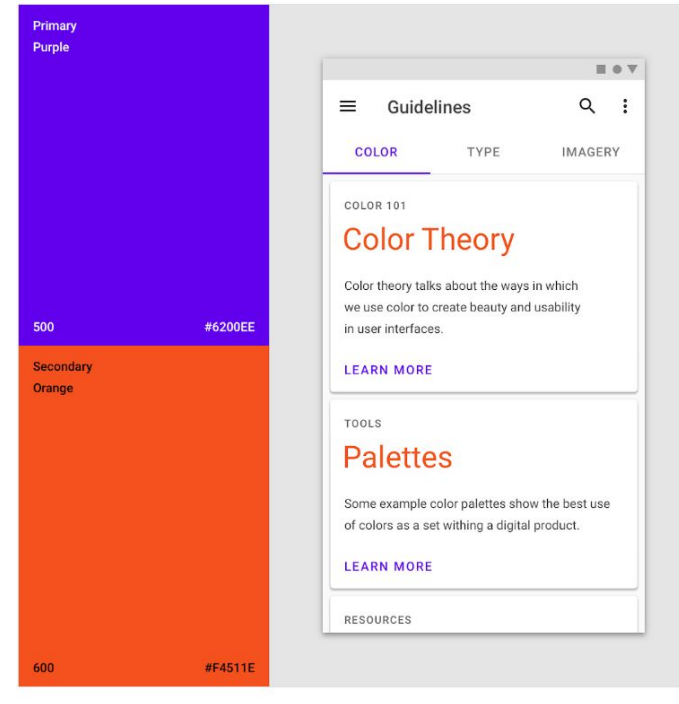
Typography and iconography

- Color can express whether text has greater, or lesser, importance relative to other text
- Color also ensures text remains legible when placed above imagery or backgrounds, which can make it difficult to read the text in front of them



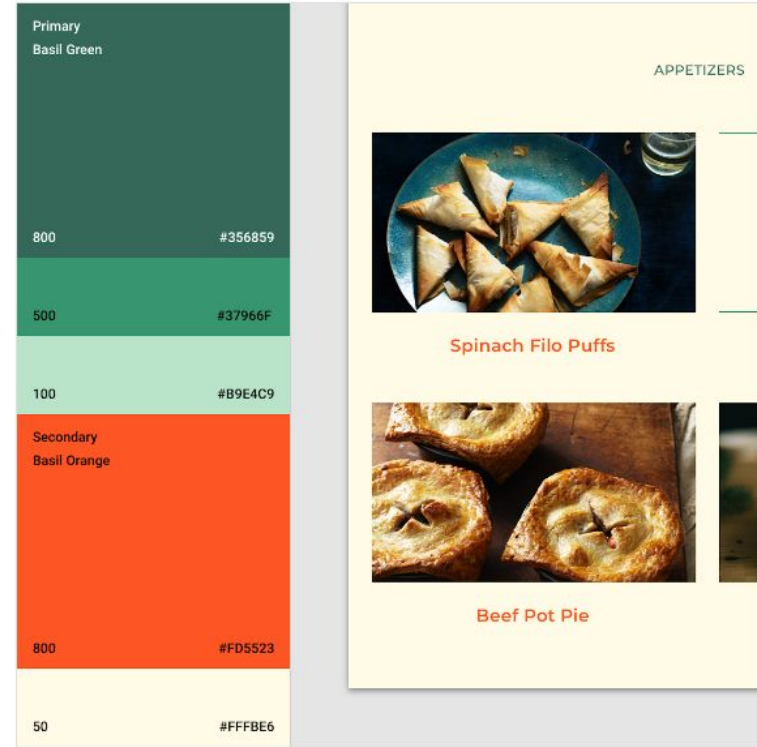
Sample colors on headlines, tabs, and buttons

- The color theme for this app consists of a primary color (purple 500) and a secondary color (orange 600)
- Orange accents the cards' headlines, and purple appears on tabs and buttons



Sample colors on headlines and tabs

- Important text, like tabs and headlines, can use your primary or secondary colors



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How do we create a color system?

- Organize all of your colors
- Define system to create extended palettes
- Establish naming conventions
- Validate new color palettes
- Define usage guidelines
- Release for testing
- Iterate
- Release to developers



Organize all of your colors

- All colors could be structured in 4 groups
 - Primary colors
 - Secondary colors
 - Neutrals
 - Semantic colors



Primary colors

- These are the colors that are most frequently used across your UI and imparts a distinct identity to the product
- These are usually the colors that a brand sets as their identity

The LinkedIn logo, featuring the word "LinkedIn" in blue with a blue square icon containing a white "in" inside the "n".

Primary color - blue



Primary color - coral



Primary color - blue

Primary colors of some renowned brands



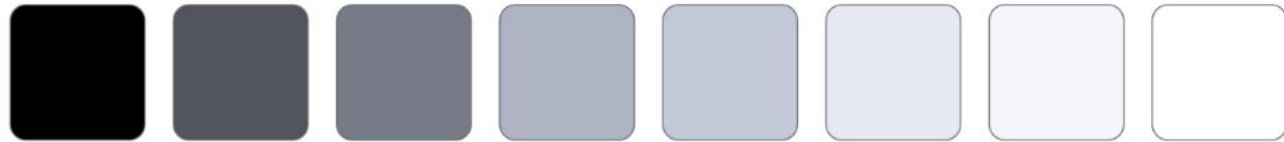
Secondary colors

- These colors highlight or complement the primary color
- These are to be used sparingly to make the UI elements stand out
- Limiting the secondary colors to three colors would bring in more recognisability and consistency



Neutrals

- These include shades of grey, all the way from White to Black
- These are used for backgrounds, text colors, etc, and form the majority of your UI



Neutrals Color Palette



Semantic colors

- These are the colors that communicate purpose
- They help users convey messages
- For example, we use Green to convey confirmation messages



Create extended palettes

- All the designers would agree that the base color is just not enough
- Once the primary, secondary and semantic colors are identified, the next step is to define a system that will allow you to generate additional shades from them



Color shades generated from different base colors



Extended palette generator

Colorbox makes creating color systems very convenient while allowing us to tweak the parameters to suit our own needs



Establish naming conventions

- Communicating using names like ‘primary color’ is easier than communicating in hardcoded values (#1F6AE3)
- Our color palettes are comprised of a base color and extended palette
- The extended palette includes the additional shades generated from the base color
- Colors are named as “Color label-XX” (for exam. primary-100)



How to name the color

- With the format 'Color label-XX'
- The color label represents the base color, such as primary, success, error instead of sea green, green, red
- How the value of XX is determined
 - Base color:100
 - Shades darker than base color: > 100
 - Shades lighter than base color: < 100



Sample color names



Naming for Primary color palette



Naming convention applied on other palettes



Validate new color palettes (1/2)

- To validate colors, identify few key UI journeys of your product
- Next, start replacing the colors in your design with newly created colors
- Check if the colors are serving the purpose they are intended to
 - For example, a secondary color must stand out to grab user attention



Validate new color palettes (2/2)

- Another important consideration is that accessibility
- Check if there is sufficient contrast between the background and text elements
- You should strive to comply with [the WCAG Color Contrast](#) or use [WebAIM: Contrast Checker](#)



Define usage guidelines

- We need to take a step ahead by defining the usage guidelines for these colors
 - Which shade should I use for secondary text?
 - What should be the background color of CTAs?
 - What colors are to be used on disabled elements?



Release for testing

- The sooner you onboard stakeholders to the design system, the better it is
- Everyone's support and inputs would make this a system that keeps evolving into a better one
- Incorporate their feedback and make iterations to the color system if required



Release to developers

- You have already built your color system and have also validated it
- The next step is to push this color system to the developers
- Now, if you are to change your primary color from sea green to sky blue, just a change in a single line of code will implement the new primary color in all the UI elements



Color Generation Tools

- Adobe color
 - <https://color.adobe.com/create/color-wheel>
- Coolors
 - <https://coolors.co/>
- Canva Color Palette Generator
 - <https://www.canva.com/colors/color-palette-generator/>



References

- <https://material.io/design/color/the-color-system.html>
- <https://material.io/design/color/applying-color-to-ui.html>
- <https://uxdesign.cc/defining-colors-in-your-design-system-828148e6210a>

